

Detection and analysis of Galileo signal in space anomalies from 2017 to 2018

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Abstract: The state of the signal in space (SIS) broadcasted by the broadcast ephemeris will affect the reliability and safety of the user navigation and positioning. Understanding the SIS anomaly of a satellite is conducive to fault allocation in advanced receiver autonomous integrity monitoring. In this study, precise ephemeris was used to calculate the SIS error of all healthy Galileo satellites from 2017 to 2018, the detection thresholds were obtained through statistics of SIS error, and the satellite SIS anomalies for two years were detected. Results show that the SIS anomalies of Galileo satellites are primarily caused by satellite clock error, the scale of which can sometimes reach tens or even hundreds of metres. Except for E24 satellite, the annual accumulation of anomalies for Galileo satellites was $< \sim 7$ h, and the average annual accumulation of anomalies is ~ 2.5 h. The accumulation of anomalies for E24 in 2017 was ~ 50 h, but this decreased to 4.5 h in 2018, indicating that its stability improved. Overall, the SIS stability in 2018 was improved compared with 2017. In addition, as the serious missing of broadcast ephemeris of Galileo satellites, it is not conducive to the continuity and reliability of the user's navigation and positioning.

1 Introduction

The European Global Navigation Satellite System Galileo entered its full operation capability (FOC) stage in December 2016, having already provided initial services. There are currently 26 satellites (four in-orbit validation (IOV) and 22 FOC satellites [1]) in orbit that can provide navigation, positioning and timing services to users around the world. During the construction of this system, the signal in space (SIS) performance of Galileo satellites improved continuously [2]. At the same time, many scholars have studied the navigation and positioning performance of the Galileo satellite at various stages [2–6]. For real-time users, the navigation and positioning performance is often related to the performance of the satellite orbits and clock errors broadcasted by the broadcast ephemeris. The offset of satellite orbit and clock data broadcasted by the broadcast ephemeris is usually known as SIS user range error or SIS range error (SISRE) [7]. SISRE can sometimes reach hundreds of thousands of metres, resulting in SIS anomalies, which affect the safety and effectiveness of users with high integrity requirements (such as advanced receiver autonomous integrity monitoring [8]). Research on SIS anomalies, from the characteristics of the SIS error to the detection methods of SIS anomalies and the causes of anomalies, first began with GPS [9–13] and then expanded to GLONASS [12, 14, 15] and the Beidou satellite navigation system [16–20]. For the Galileo satellite, the SISRE accuracy of the IOV satellite was evaluated [13, 21], which was initially ~ 1 – 2 m. The error distribution of the orbit, clock and SISRE for Galileo satellite in 2015 was analysed [22]. The relevant calculation methods of SISRE were summarised and the SISRE of multi-GNSS was evaluated [23], the RMS SISRE of Galileo satellite in 2017 was found to be ~ 0.2 m, and the performance of the system declined three times during analysis [23]: the first time, the update rate of navigation ephemeris delay [24]; the second time is a satellite clock frequency jump in the SIS results when the accuracy reached 490 m [24]; and the third time is a small satellite clock frequency jump that made the broadcast ephemeris clock offset error ~ 30 m. Long-term analysis has revealed that the

average RMS of SISRE for Galileo satellites were 0.58 m (2015), 0.29 m (2016), 0.23 m (2017) and 0.22 m (2018) [1].

The latest typical case occurred on 11–18 July 2019. At this time, due to the delayed update of the broadcast ephemeris, the Galileo system experienced a process from service downgrade to full interruption [25], and the service recovery caused close attention from all parties. Therefore, it is necessary to study the SIS anomalies of the Galileo satellites. Although the Galileo broadcast ephemeris currently provides the SIS accuracy (SISA (m)), we counted the SISA in the broadcast ephemerides and found that they did not change with time, so it is not conducive to use SISA to calculate the threshold of anomaly detection similar to the SIS anomaly detection of GPS [12]. In this study, when the precise ephemeris is used to detect the SIS anomalies, the detection threshold of the satellite is obtained based on the statistics of SISRE. The SISRE of the Galileo satellites was then calculated for the period of 2017–2018, using broadcast ephemeris and precision ephemeris. In addition, the Galileo SIS anomalies for two years were detected through gross error elimination, threshold calculation, and anomaly detection and some satellites were selected to analyse the causes of the anomaly in Section 3.4.

2 Principle

2.1 Calculation of SIS error

SISRE refers to the deviation between the satellite orbit and clock error broadcasted by satellite broadcast ephemeris and their true values. In statistics, the global average SISRE of the satellite orbit and clock error in line-of-sight to the users is typically used; the main formula used to calculate this is as follows:

$$\text{SISRE} = \sqrt{(S_1 * R - c * T)^2 + (A^2 + C^2) / S_2} \quad (1)$$

where R , A and C represent satellite orbit errors in radial, along-track and cross-track directions; T represents the satellite clock error, c is the speed of light and S_1 and S_2 represent the projection coefficient factors of the errors.

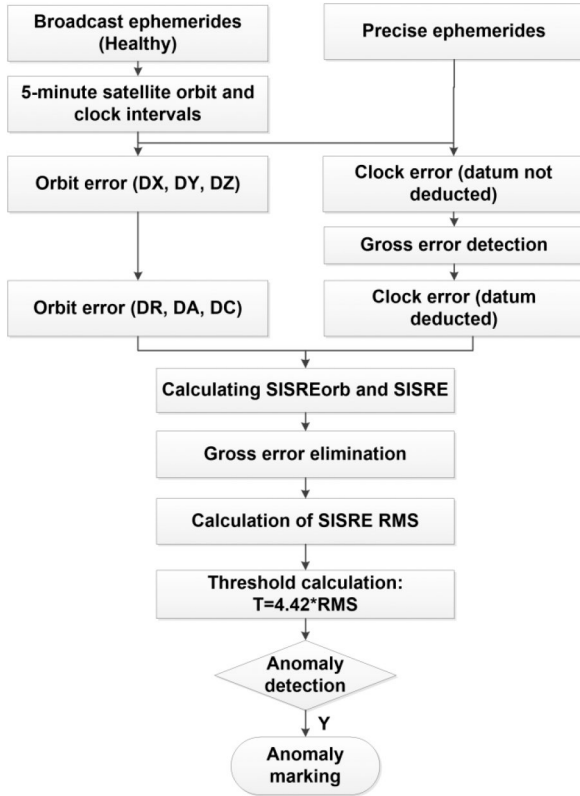


Fig. 1 Detection flow of SIS anomaly for the Galileo Satellite

If only the impact of satellite orbit errors on SISRE is evaluated, then

$$\text{SISRE}_{\text{orb}} = \sqrt{(S_1 * R)^2 + (A^2 + C^2)/S_2} \quad (2)$$

Usually, the projection coefficient factor is set to a fixed value [12]. For Galileo satellites S_1 and S_2 , $S_1 = 0.984$ and $S_2 = 61.2$ [12, 13].

Usually, the precise orbit and clock error products provided by the International GNSS Service (IGS) [26] can be considered as true value, as the precision of them is 2–3 cm, which can be used as a true value reference [12]. However, the unification of the space and time reference datum should be considered, for several reasons. First, the difference between the precise ephemeris and broadcast ephemeris coordinate systems is ignored. Second, the reference time of the broadcast ephemeris is unified to be consistent with the precise ephemeris. Third, as the antenna reference points of broadcast ephemeris and precise ephemeris are different [27], it is necessary to correct the antenna phase centre. Here, the antenna phase centre parameters provided by Galileo satellite metadata (GSA) were selected for correction [28]. The fourth consideration is the unification of the satellite clock reference datum.

Then, the calculation process for SISRE is as follows: first, the satellite orbit and clock error are calculated at equal intervals with the precise ephemeris, using the broadcast ephemeris. Second, the difference in the satellite orbit and clock error between the broadcast ephemeris and precise ephemeris is calculated. Third, the satellite orbit error is transferred from the Earth-centred Earth-fixed coordinate system to the satellite orbit coordinate system. In the fourth step, the clock error reference datum is subtracted, and finally the orbit error, clock error, SISREorb and SISRE of the satellite are obtained.

In the above calculation process, the unification of satellite clock reference datum is as follows:

The difference, dt^i , between the i th satellite broadcast ephemeris clock error, t^i and the precise ephemeris clock error, T^i , is expressed as

$$dt^i = t^i - T^i \quad (3)$$

The broadcast ephemeris and precise ephemeris each use a different time datum to calculate satellite clock error, and this difference causes the same systematic deviation for all satellites in the same epoch and the same system. Therefore, when calculating the satellite clock error in this study, the average clock error method is used to subtract the clock error datum as

$$\Delta t^i = dt^i - \mu \quad (4)$$

where $\mu = \sum_{i=1}^n dt^i/n$ means taking the average value of dt^i for all satellites and Δt^i is the final clock error.

When the satellite clock error datum is subtracted, the clock error data should be pre-processed, to remove the effect of the error satellite on the calculation of the clock error datum, μ [19].

2.2 Detection of SIS anomaly

Based on the above calculation process, the designed Galileo satellite SIS anomaly detection process is as follows:

(i) *The sampling interval is set to 5 min:* As the update interval of Galileo broadcast ephemeris is 10 min and the precise ephemeris sampling interval is 5 min, in order to ensure sufficient sampling data, the interval is set to 5 min. At the same time, in order to reduce the impact of the ephemeris update delay, the broadcast ephemeris fitting time should meet $|t - t_{\text{oc}}| \leq 10$ min.

(ii) *The satellite orbit and clock error are calculated:* A healthy broadcast ephemeris (health status flag is 0) is selected to calculate the satellite orbit and clock error and the broadcast ephemeris is compared with the precise ephemeris to get the corresponding orbit and clock error, paying attention to the unification of space and time reference datum.

(iii) *SIS error calculation:* The SISREorb and SISRE of the satellite are calculated, where SISRE is used as input data for the detection of SIS anomalies.

(iv) *Gross error elimination:* For Galileo satellites, the 3-m threshold is selected to remove the large gross error in the SISRE, and the filtered data are used for the next stage of the detection threshold calculation.

(v) *The RMS of the SISRE is calculated for each satellite:* The trim data where $\alpha = 0.001$ from the result of step (4) is used to calculate the RMS of SISRE for each satellite.

(vi) *Anomaly detection:* Based on the assumption of the normal distribution of errors, according to the judgment condition of formula (5), the detection threshold is set to $T = 4.42 \times \text{RMS}$ and satellite anomalies in the original SISRE sequence are detected

$$P(|\text{SISRE}| > 4.42 \times \text{RMS}) = 10^{-5} \quad (5)$$

The overall flow of the Galileo satellite SIS anomaly detection designed in this work is shown in Fig. 1.

3 Results analysis

3.1 Data sources

The daily broadcast ephemeris of the multi-GNSS experiment (MGEX) and the precise ephemeris of Helmholtz-Centre Potsdam-German Research Centre for Geosciences (GFZ) from 1 January 2017 to 1 January 2018 were selected. The sampling interval for which the SISRE was calculated by Galileo satellite is 5 min, which is consistent with the precise ephemeris.

3.2 Valid data of broadcast ephemeris Galileo satellite

According to the satellite health status indicators in the Galileo satellite broadcast ephemeris, the data sequences from 2017 to 2018 were plotted using Matlab software, as shown in Fig. 2.

In Fig. 2, Blue dots indicate that healthy broadcast ephemerides were collected during the valid period. Red dots indicate that unhealthy broadcast ephemerides were collected during the valid period. The interrupted portion indicates that broadcast ephemerides were lacking.

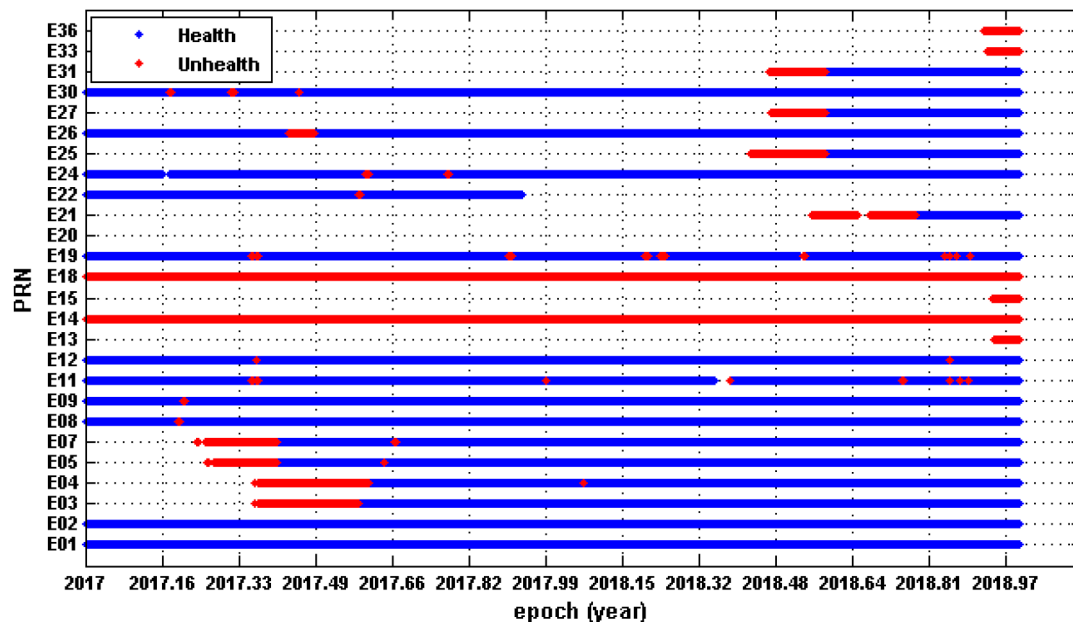


Fig. 2 Health status of Galileo satellites from 2017 to 2018

Table 1 Statistics of effective Galileo satellites in 2017 and 2018

Year	Satellites
2017	E01, E02, E03, E04, E05, E07, E08, E09, E11, E12, E19, E22, E24, E26, E30
2018	E01, E02, E03, E04, E05, E07, E08, E09, E11, E12, E19, E21, E24, E25, E26, E27, E30, E31

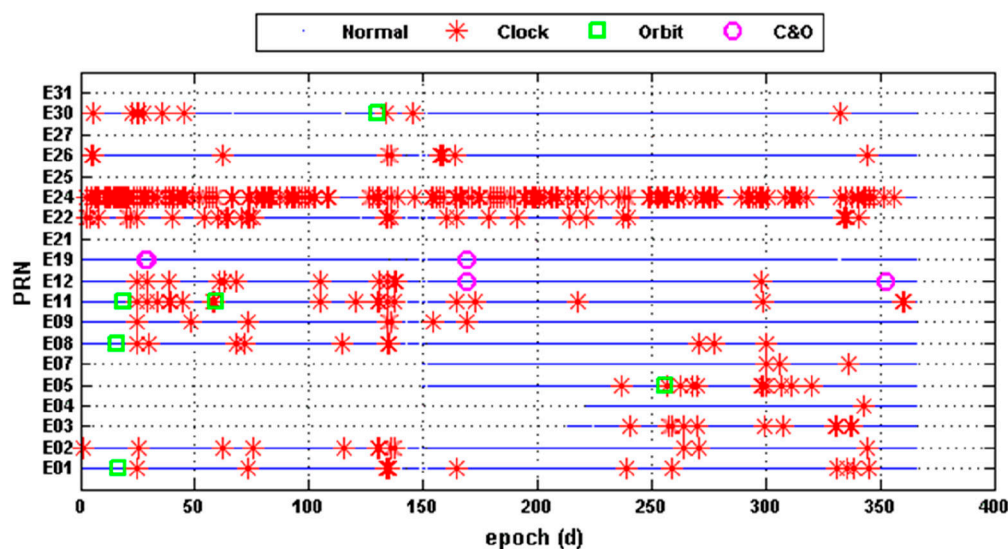


Fig. 3 SIS anomalies of each Galileo satellite by epoch (day) in 2017

As shown in Fig. 2, the health statuses of the E13, E14, E15, E18, E33 and E36 satellites were marked as either unhealthy or debugged, so there were 19 effective satellites in total, and the statistics of satellites in 2017 and 2018 are shown in Table 1. Subsequent analysis of the 19 satellites will be carried out.

As shown in Table 1, there were 15 satellites in 2017 and 18 satellites in 2018, respectively. Among them, satellite E22 only has data in 2017; satellites E21, E25, E27 and E31 began to provide data in 2018.

3.3 Statistical results of Galileo satellite anomalies

The SIS anomalies of the Galileo satellites in 2017 and 2018 were detected using the method and process shown in Fig. 1, and the statistical results of the anomaly detection were as shown in Figs. 3 and 4, respectively.

In Figs. 3 and 4, the blue dots indicate normal SISRE. The red asterisks, green squares and purple circles indicate anomalies

caused by satellite clock errors, satellite orbit errors and a combination of both satellite clock and orbit errors (recorded as C&O), respectively. The interrupted portions indicate missing broadcast ephemeris or precise ephemeris.

As shown in Figs. 3 and 4, all of the Galileo satellites inevitably have SIS anomalies. Compared to 2017, the satellite's SIS anomalies were significantly reduced in 2018. According to the classification of anomalies, it can be seen that the SIS anomalies of the Galileo satellite are mainly caused by satellite clock errors. Among them, the E24 satellite had the most anomalies in 2017 and 2018, and there were several satellites that had anomalies on one day in 2018. The anomaly numbers of the E21, E27 and E31 were unusually rare, as these satellites were newly added to the service in 2018.

Each point in Figs. 3 and 4 represent a 5-min anomaly. The cumulative duration of the Galileo satellite SIS anomalies in 2017 and 2018 was calculated, as shown in Table 2.

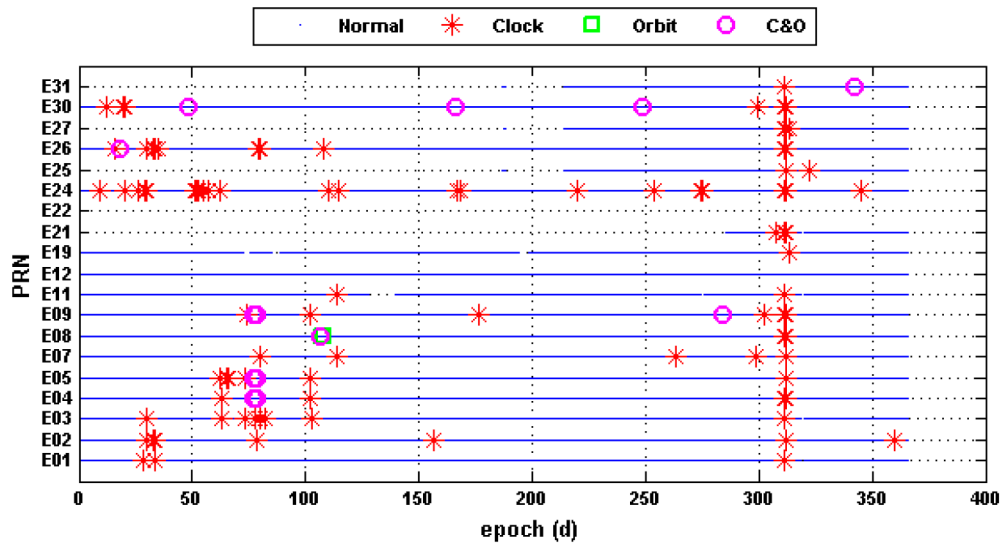


Fig. 4 SIS anomalies of each Galileo satellite by epoch (day) in 2018

Table 2 Anomaly statistics of Galileo satellite in 2017–2018 (unit: hour)

PRN	2017			2018		
	Clock	Orbit	C&O	Clock	Orbit	C&O
E01	3.42	0.17	0.00	0.92	0.00	0.00
E02	3.67	0.00	0.00	1.58	0.00	0.00
E03	2.83	0.00	0.00	1.42	0.00	0.00
E04	0.08	0.00	0.00	1.33	0.00	1.08
E05	1.83	0.33	0.00	1.25	0.00	3.00
E07	0.50	0.00	0.00	1.17	0.00	0.00
E08	2.50	0.17	0.00	1.17	2.17	1.25
E09	1.42	0.00	0.00	1.67	0.00	4.92
E11	5.75	0.50	0.00	0.33	0.00	0.00
E12	2.67	0.00	1.50	0.00	0.00	0.00
E19	0.00	0.00	0.33	0.17	0.00	0.00
E21	—	—	—	0.83	0.00	0.00
E22	5.50	0.00	0.00	0.00	0.00	0.00
E24	49.42	0.00	0.00	4.50	0.00	0.00
E25	—	—	—	0.50	0.00	0.00
E26	3.42	0.00	0.00	2.67	0.00	3.00
E27	—	—	—	2.00	0.00	0.00
E30	2.17	0.33	0.00	2.25	0.00	1.92
E31	—	—	—	0.33	0.00	3.33

‘C&O’ represents the anomalies caused by a combination of both orbit and clock errors

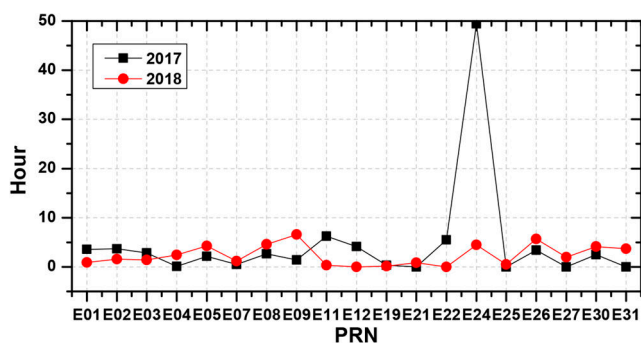


Fig. 5 Total hours of satellite anomalies for Galileo satellites in 2017–2018

As shown in Table 2, the SIS anomaly of E24 in 2017 was ~49.42 h, which is much larger than that of the other satellites. The total accumulation hours of the satellite anomaly is shown in Fig. 5.

As shown in Fig. 5, except for the E24 satellite (~50 h in 2017), the total hours of the Galileo satellites’ anomalies were no more

than 7 h, and the annual average cumulative anomalies of each satellite was ~2.5 h.

3.4 Anomaly analysis of typical satellites

For the Galileo satellite anomalies detected in Section 3.3, some satellite anomalies are selected here for detailed analysis.

3.4.1 Satellite E01 on day of year (DOY) 134 of 2017: The statistics of the satellite’s SISRE, clock error, health status and satellite clock error sequences are shown in Fig. 6.

In Fig. 6, Dclk (a) indicates the clock differences between broadcast and precise ephemerides deducted the clock datum and satellite clock sequences (b) are the original satellite clock data obtained by broadcast and precise ephemerides. Similar representations in the following figures will not be explained.

As shown in Fig. 6, the anomalies of the E01 satellite after 1 h on DOY 134 of 2017 were mainly caused by satellite clock jumps (Fig. 6b). At the same time, the precise ephemeris of the E01 satellite was missing at 09:45–09:50 and at 10:45–10:50. This indicates that there may be anomalies in the satellite clock here. It can be seen from the statistical sequence of Fig. 6a that the data

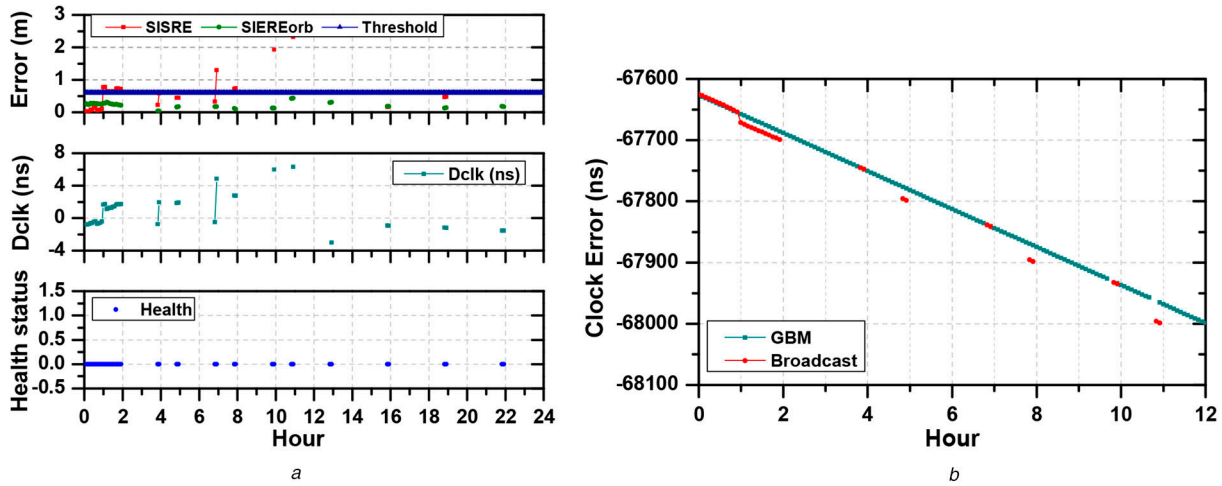


Fig. 6 Anomalies in satellites E01 on DOY 134 of 2017

(a) Sequences of SISRE, SISREorb, clock error and health status, (b) Satellite clock sequences from 0 to 12 h

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E01 2017 05 14 01 50 00 -6.769667379558E-05-9.137579581875E-12-3.469446951954E-18
1.100000000000E+01 1.243125000000E+02 2.568678424338E-09 4.705321707993E-01
5.869194865227E-06 1.805867068470E-04 2.764165401459E-06 5.440617164612E+03
6.600000000000E+03 7.450580596924E-09 1.412535258152E+00 3.725290298462E-09
9.976141831151E-01 2.960312500000E+02 1.981389136331E+00-5.582018227751E-09
3.192990143713E-10 5.170000000000E+02 1.949000000000E+03
3.120000000000E+00 0.000000000000E+00-5.587935447693E-09-6.519258022308E-09
7.266000000000E+03
E01 2017 05 14 03 50 00 -6.774504436180E-05-8.682832230988E-12 0.000000000000E+00
1.320000000000E+02 1.267187500000E+02 2.564392531528E-09 1.370877460400E+00
5.973502993584E-06 1.798031153157E-04 3.449618816376E-06 5.440613954544E+03
1.380000000000E+04 2.235174179077E-08 1.412495103976E+00 2.235174179077E-08
9.976166071703E-01 2.839062500000E+02 1.973662904193E+00-5.560588763699E-09
3.135844906242E-10 2.580000000000E+02 1.949000000000E+03
3.120000000000E+00 0.000000000000E+00-5.587935447693E-09 0.000000000000E+00
1.384000000000E+04
E01 2017 05 14 04 50 00 -6.779574323446E-05-9.208633855451E-12-3.469446951954E-18
1.380000000000E+02 1.392187500000E+02 2.51153186867E-09 1.808794782557E+00
6.552785634995E-06 1.791402464733E-04 3.490597009659E-06 5.440617284775E+03
1.740000000000E+04-9.313225746155E-09 1.412475215605E+00 4.842877388000E-08
9.976178623540E-01 2.830000000000E+02 1.982055922688E+00-5.537016353242E-09
2.928693420408E-10 5.170000000000E+02 1.949000000000E+03
3.120000000000E+00 0.000000000000E+00-5.587935447693E-09-6.519258022308E-09
1.740600000000E+04
E01 2017 05 14 06 50 00 -6.783875869587E-05-8.682832230988E-12 0.000000000000E+00
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Fig. 7 Part of the broadcast ephemerides of E01 on DOY 134 of 2017

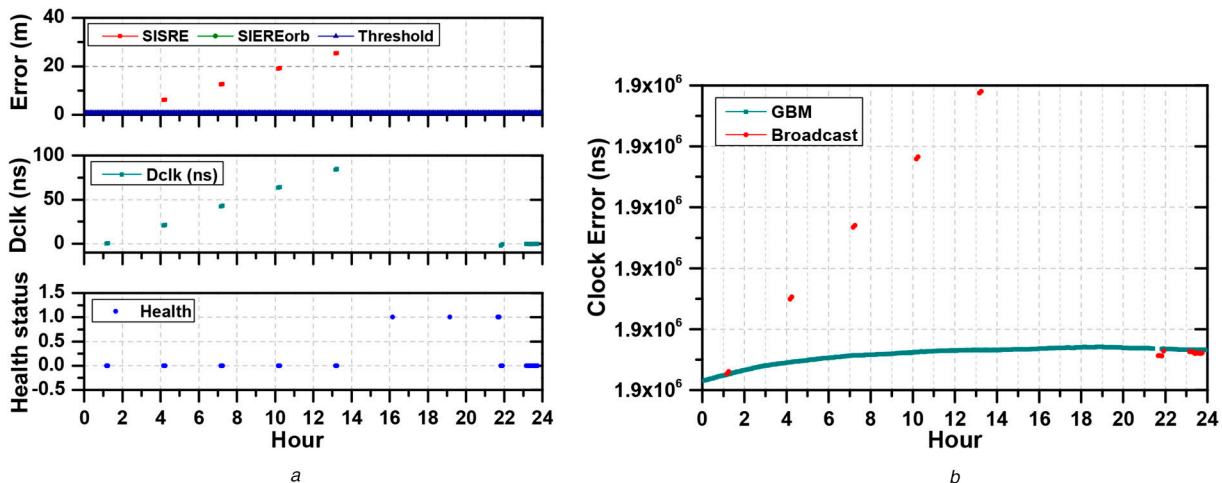


Fig. 8 Anomalies in satellite E08 on DOY 134 of 2017

(a) Sequences of SISRE, SISREorb, clock error and health status, (b) Satellite clock sequences

sequence after 2 o'clock on that day has only a few epochs, with many interruptions. The following is a partial broadcast ephemeris, as shown in Fig. 7.

The broadcast ephemeris of the E01 satellite from 01:50–06:50 on DOY 134 of 2017 was intercepted, as shown in Fig. 7. The ephemeris of the E01 satellite has only individual epochs; the rest are missing. Therefore, the absence of statistical sequences is mainly related to the absence of broadcast ephemeris.

3.4.2 Satellite E11 on DOY 360 of 2017: The statistics of the E11 satellite's SISRE, clock error, health status and satellite clock error sequences are shown in Fig. 8.

As shown in Fig. 8, the E11 satellite experienced a SIS anomaly of ~35 m in SISRE between 04 and 14 h on DOY 360 of 2017. This was mainly caused by the satellite clock jump (Fig. 8b). The healthy status of the broadcast ephemeris lasted between 16:00 and

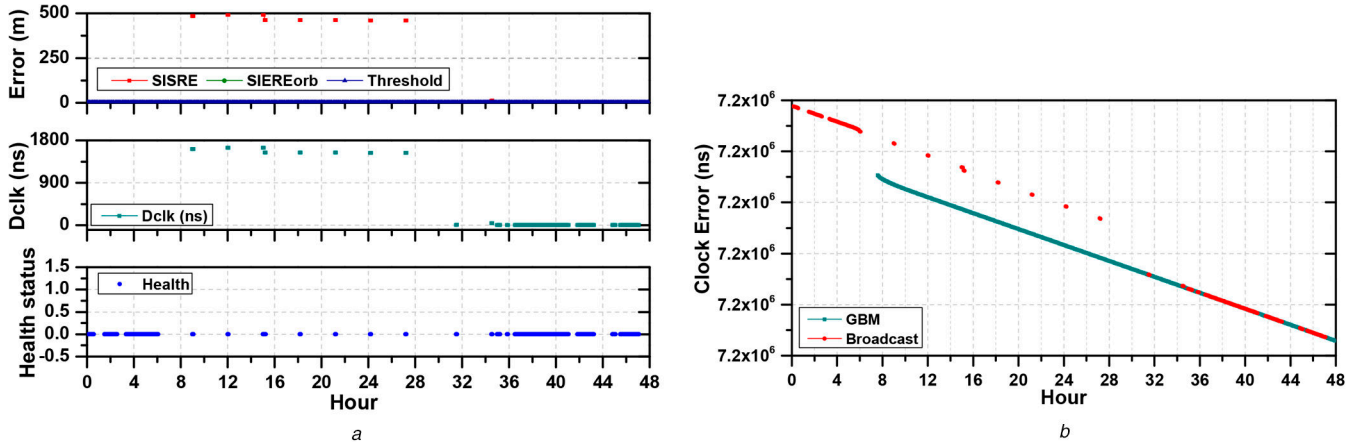


Fig. 9 Anomalies in satellite E26 from DOY 157 to 158 of 2017
(a) Sequences of SISRE, SISREorb, clock error and health status, (b) Satellite clock sequences

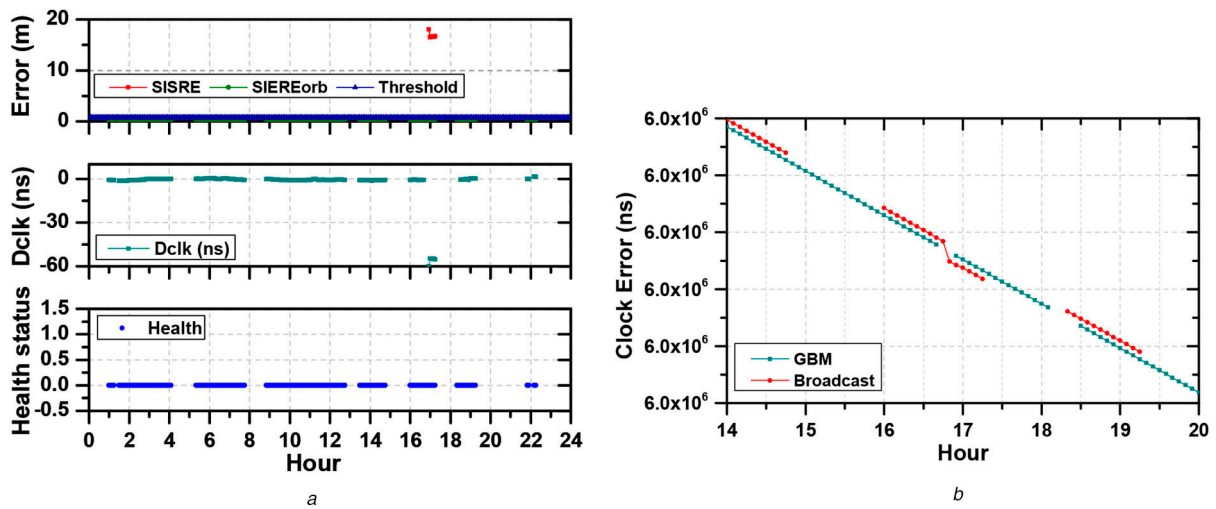


Fig. 10 Anomalies in satellite E26 on DOY 107 of 2018
(a) Sequences of SISRE, SISREorb, clock error and health status, (b) Satellite clock sequences from 14 to 20 h

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E26 2018 04 17 17 10 00 6.039245752618E-03-4.501998773776E-11 0.000000000000E+00
7.000000000000E+00 7.625000000000E+00 2.647253125861E-09-2.048951378958E+00
2.346932888031E-07 1.982636749744E-04 1.271255314350E-05 5.440619007111E+03
2.346000000000E+05 4.842877388000E-08-2.436618667611E+00-3.539025783539E-08
9.748565408593E-01 7.484375000000E+01-1.183815620707E+00-5.317721504445E-09
4.375182243902E-10 5.170000000000E+02 1.997000000000E+03
-1.000000000000E+00 0.000000000000E+00 2.095475792885E-09 2.561137080193E-09
2.352660000000E+05
E26 2018 04 17 18 20 00 6.039122527000E-03-4.314415491535E-11 0.000000000000E+00
1.400000000000E+01-1.075000000000E+01 2.743685714094E-09-1.541777197666E+00
-4.600733518600E-07 1.978327054530E-04 1.305527985096E-05 5.440613519669E+03
2.388000000000E+05 5.029141902924E-08-2.436640949316E+00-1.303851604462E-08
9.748584631337E-01 6.562500000000E+01-1.170296382193E+00-5.301649406407E-09
4.303750697063E-10 5.170000000000E+02 1.997000000000E+03
3.120000000000E+00 0.000000000000E+00 2.095475792885E-09 2.561137080193E-09
2.394660000000E+05
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Fig. 11 Part of broadcast ephemerides of E26 on DOY 107 of 2018

22:00, then the broadcast ephemeris is marked as being non-healthy, and then the normal ephemeris recovered.

3.4.3 Satellite E26 from DOY 157 to 158 of 2017: The statistics of the E26 satellite's SISRE, clock error, health status and satellite clock error sequence are shown in Fig. 9.

As shown in Fig. 9, the E26 satellite produced a continuous SIS anomaly of ~500 m in SISRE from DOY 157–158 of 2017, lasting from 8 h for DOY 157 to 04 h for DOY 158. The statistics of the broadcast ephemeris and the precise ephemeris clock error show that the anomaly is mainly caused by the satellite clock jump (Fig. 9b). As the broadcast ephemeris is a forecast ephemeris, it cannot respond in time. Subsequently, the broadcast ephemeris was missing from 06–08 h, and it returned to normal after DOY 158, after ~09 h.

3.4.4 Satellite E26 on DOY 107 of 2018: The statistics of the E26 satellite's SISRE, clock error, health status and satellite clock error sequence are shown in Fig. 10.

As shown in Fig. 10, the E26 satellite generated a SIS anomaly caused by the satellite clock jump at 16: 55–17: 15 on DOY 107 of 2018 (Fig. 10b), resulting in the SISRE reaching nearly 20 m. Then the ephemeris was missing, and part of the broadcast ephemeris of the corresponding period was intercepted, as shown in Fig. 11.

As shown in Fig. 11, the broadcast ephemeris was missing at 17:20–18:10, and was recovered at 18:20. Combined with Fig. 10, it can be seen that the recovered ephemeris provided a normal SIS.

4 Conclusion

Using the broadcast ephemeris provided by the MGEX, and the precise ephemeris provided by GFZ, the SIS errors of the Galileo

satellites were calculated over two years (2017–2018), and the satellite SIS anomalies were detected. The causes of the anomalies reveal that the SIS anomalies of the Galileo satellites are mainly related to the satellite clock error, which causes the SISRE to reach tens or hundreds of metres. The anomalies in 2018 were significantly reduced, compared to those in 2017, indicating an improvement in the performance of the SIS. Except for the E24 satellite, the annual average cumulative hours of the Galileo satellite anomalies was within ~ 7 h, and the annual average cumulative anomaly of each satellite was ~ 2.5 h. In addition, in 2017 and 2018, there were several satellites that experienced abnormalities at the same time. Detailed analysis revealed that there were frequently ephemerides missing in the broadcast ephemeris broadcasted by the Galileo satellites, which is not conducive to the user's navigation and positioning, and needs to be further improved.

5 Acknowledgments

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